

Discrete Berth Allocation with a Shared Navigation Channel, Vessel–Berth Compatibility, and Priority Berth Reservations

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Abstract

We schedule one week of vessel calls at the Port of San Antonio. Each vessel is assigned to a *compatible* berth and a start time; a single *navigation channel* (a shared one-vessel-at-a-time waterway) must be transited once to enter and once to exit; and a subset of priority *service* vessels hold fixed berth-time *reservations* that are kept exclusively for them. The model is solved for a chosen operational objective: total *waiting time* or total *berth idle time*. Objectives are unweighted (vessel priority weights are deferred). The service time τ that drives the schedule of the non-reserved vessels is supplied by an upstream predictor.

1 Sets, parameters, and variables

Symbol	Meaning
$\mathcal{I} = \{1, \dots, N\}$	vessels in the planning week
$\mathcal{B} = \{1, \dots, B\}$	berths
$\mathcal{B}_i \subseteq \mathcal{B}$	berths <i>compatible</i> with vessel i
$\mathcal{S} \subseteq \mathcal{I}$	priority “service” vessels (fixed reservations)
$\mathcal{E} = \{(i, \text{in}), (i, \text{out}) : i \in \mathcal{I}\}$	the $2N$ channel transit events
$a_i \geq 0$	arrival time of vessel i
$\tau_i > 0$	berth-occupation time: <i>predicted</i> for $i \notin \mathcal{S}$, fixed at D_r for $r \in \mathcal{S}$
$b_r \in \mathcal{B}$	reserved berth of service vessel $r \in \mathcal{S}$ (fixed input)
$R_r \geq 0$	reserved start time of service vessel r (fixed input)
$D_r > 0$	reserved (committed) berth-occupation duration of r (fixed input)
$c \geq 0$	channel transit time (enter or exit)
M	big- M constant ($\geq$ horizon + max berth workload)
$x_{ib} \in \{0, 1\}$	1 if vessel i uses berth b (only for $b \in \mathcal{B}_i$)
$s_i \geq a_i$	berth start time of vessel i
$z_{ijb} \in \{0, 1\}$	1 if i precedes j at berth b
$e_i^{\text{in}} \geq a_i$	inbound (entry) channel-transit start
$e_i^{\text{out}} \geq 0$	outbound (exit) channel-transit start
$y_{ef} \in \{0, 1\}$	1 if transit event e precedes event f
$LC_b, FS_b \geq 0$	last completion / first start at berth b (idle objective)

Table 1: Notation. Departure of vessel i is $e_i^{\text{out}} + c$; transit event (i, k) occupies the channel over $[t_i^k, t_i^k + c]$ with $t_i^{\text{in}} = e_i^{\text{in}}$ and $t_i^{\text{out}} = e_i^{\text{out}}$.

2 Objective (selectable)

Service vessels are pinned by their reservations (§below), so the optimization acts on the remaining non-service vessels $\mathcal{I} \setminus \mathcal{S}$. The model is solved for one of two unweighted operational objectives:

$$\text{(Waiting)} \quad \min \sum_{i \in \mathcal{I} \setminus \mathcal{S}} (e_i^{\text{in}} - a_i), \quad (1)$$

$$\text{(Idle)} \quad \min \sum_{b \in \mathcal{B}} (LC_b - FS_b). \quad (2)$$

Waiting time (1) is the total time the non-service vessels wait at anchorage before they may enter the channel; the service duration cancels, so only the controllable pre-berth delay is penalized. *Berth idle time* (2) is summed over all berths, so a berth's occupied span includes any reserved blocks on it; since the total occupation $\sum_i \tau_i$ is a constant, minimizing idle time is equivalent to minimizing the summed per-berth occupied span $\sum_b (LC_b - FS_b)$, and the true idle KPI is this span minus $\sum_i \tau_i$.

3 Constraints

Assignment and arrival.

$$\sum_{b \in \mathcal{B}_i} x_{ib} = 1 \quad \forall i \in \mathcal{I} \quad (3)$$

$$e_i^{\text{in}} \geq a_i, \quad s_i \geq a_i \quad \forall i \in \mathcal{I} \setminus \mathcal{S} \quad (4)$$

Constraint (3) assigns each vessel to exactly one *compatible* berth; the variables x_{ib} exist only for $b \in \mathcal{B}_i$, so incompatible assignments are structurally excluded (for a service vessel r we set $\mathcal{B}_r = \{b_r\}$, so this constraint pins it to its reserved berth). Constraint (4) is the dynamic arrival release: a non-service vessel may neither enter the channel nor begin berth service before it has arrived ($e_i^{\text{in}} \geq a_i$ and $s_i \geq a_i$). Service vessels are exempt—their timing is fixed by the reservation (12), so a reserved block is held even if the vessel arrives earlier or later than R_r .

Navigation channel.

$$s_i \geq e_i^{\text{in}} + c \quad \forall i \in \mathcal{I} \quad (5)$$

$$e_i^{\text{out}} \geq s_i + \tau_i \quad \forall i \in \mathcal{I} \quad (6)$$

$$t_f \geq t_e + c - M(1 - y_{ef}) \quad \forall e, f \in \mathcal{E}, e < f, v(e) \neq v(f) \quad (7)$$

$$t_e \geq t_f + c - M y_{ef} \quad \forall e, f \in \mathcal{E}, e < f, v(e) \neq v(f) \quad (8)$$

A vessel may berth only after its inbound transit finishes (5), and may begin its outbound transit only after service completes (6). Constraints (7)–(8) are the channel mutual-exclusion disjunction: for every pair of transit events of *different* vessels ($v(e)$ is event e 's vessel), one transit must clear the channel before the other starts—so the channel holds exactly one vessel at a time, in either direction. The same-vessel pair $(i, \text{in}), (i, \text{out})$ is already ordered by (5)–(6). Setting $c = 0$ collapses the transit constraints, recovering the berth-only model.

Berth sequencing (no overlap).

$$z_{ijb} + z_{jib} \leq 1 \quad \forall b, i < j : b \in \mathcal{B}_i \cap \mathcal{B}_j \quad (9)$$

$$z_{ijb} + z_{jib} \geq x_{ib} + x_{jb} - 1 \quad \forall b, i < j : b \in \mathcal{B}_i \cap \mathcal{B}_j \quad (10)$$

$$s_j \geq s_i + \tau_i - M(1 - z_{ijb}) \quad \forall (i, j, b), i \neq j : b \in \mathcal{B}_i \cap \mathcal{B}_j \quad (11)$$

Constraint (9) permits at most one ordering direction for a pair at berth b . Constraint (10) forces an ordering whenever both vessels are assigned to b : its right-hand side equals 1 only when $x_{ib} = x_{jb} = 1$, and is non-positive (vacuous) otherwise. The big- M precedence (11) then keeps the later vessel from starting until the earlier one finishes, so their berth occupations cannot overlap. Precedence triples and z variables are created only for pairs that can share a berth ($b \in \mathcal{B}_i \cap \mathcal{B}_j$).

Priority berth reservations.

$$s_r = R_r, \quad \tau_r = D_r \quad \forall r \in \mathcal{S} \quad (12)$$

Each service vessel r carries a reservation (b_r, R_r, D_r) given as fixed input. Together with $\mathcal{B}_r = \{b_r\}$ in (3), constraint (12) pins r to its reserved berth, start time, and occupation duration, so r holds the interval $[R_r, R_r + D_r]$ on berth b_r . Exclusivity needs no extra constraint: the no-overlap sequencing (9)–(11), applied between r and any other vessel eligible for berth b_r , keeps that interval empty of all other vessels—the slot is *left empty*, with no backfilling, regardless of whether r arrives before or after R_r . The reservation blocks only the berth: service vessels still transit the shared channel like any other vessel ((5)–(8)).

Berth idle-time linearization (only for objective (2)).

$$LC_b \geq s_i + \tau_i - M(1 - x_{ib}) \quad \forall i, b \in \mathcal{B}_i \quad (13)$$

$$FS_b \leq s_i + M(1 - x_{ib}) \quad \forall i, b \in \mathcal{B}_i \quad (14)$$

$$FS_b \leq LC_b \quad \forall b \in \mathcal{B} \quad (15)$$

Under minimization, LC_b is driven down to the latest completion and FS_b up to the earliest start among vessels assigned to berth b ; the big- M switches each bound off for vessels not at b . Constraint (15) caps an empty berth's span at zero.

4 Implementation pointers

The MILP is `DiscreteBAP` in `optimizers/src/bap_optim/discrete_bap.py`, with constructor argument `objective ∈ {"waiting", "idle"}`. The instance descriptor `BAPInstance` (in `bap_optim/instance.py`) carries `arrivals`, `berth_compat`, `channel_time`, and the service reservations as fixed (b_r, R_r, D_r) inputs. *Note*: the code still implements the earlier no-wait window (the `service / latest_start` fields and `hard_windows` flag); migrating it to the reservation model of (12) is a pending change.